Big Data Analysis for Healthcare Application using Minhash and Machine Learning in Apache Spark Framework

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***Abstract*—Analysing data on a large scale is becoming important and engages in convincing many researchers to use new platforms and tools that can handle large amounts of data. In this article, we present new evaluation sentiment analysis for large-scale datasets of COVID-19 Vaccine Stance tweets and COVID-19 Tweets IEEE data port datasets in the Apache Spark data system. The Apache Spark Scalable Machine Learning Library (ML) is used. We designed hybrid minhash models from the library with four classification methods: Logistic Regression (LR), Naive Bayes, Support Vector Machine and Random Forest classifiers in a parallel and distributed manner. In addition, Minhash with locality Sensitive hashing (Minhash-LSH) is compared to Minhash-ML. Performance parameters such as user, system and real time, time consumed, and accuracy have been applied in the comparative analysis to analyse the behaviour of the classifiers in the AWS spark Cluster, Local Spark cluster and in conventional system. Results have indicated that the models in spark environment was extremely effective for processing large-dimension data, which cannot be processed with conventional implementation or take much time related to some algorithms. The proposed model achieves accuracy above 99% in case of Vaccine tweet dataset when classified with Minhash-RF and Minhash-LR classifiers. Also, 100% in case of COVID-19 Tweets Provided by IEEE data port when using Minhash-SVM, Minhash-RF and Minhash-LR classifiers.**

Keywords— Big data, Apache spark, Sentiment, Machine Learning, Minhash-LSH, COVID-19 Tweets, Healthcare.

I. Introduction

Big data analytics has gained traction in a variety of industries, with the healthcare industry emerging as one of the most promising [1]. The three distinctive properties of big data, namely Volume, Velocity, and Variety, each have their unique characteristics in the vast amounts of healthcare data [2].

The big data sources of healthcare system are divided into three categories: structured data (data that adheres to a set of data types, formats, and structures), semi structured (data having a simple structure and self-descriptive nature) or unstructured format (data that is devoid of any intrinsic structure). This becomes an issue in data analysis processes. The growing requirement to manage ever-increasingly large data sets has sparked a lot of interest in big data tools development. When applying any algorithm on enormous datasets, the requirement for big data analytics frameworks becomes clear [3].

In a local system, any such process requires a single core of the CPU. A framework that makes use of the current CPU in the local system in a distributed context is required. Hadoop is one of the most well-known solutions for accomplishing this goal. Hadoop architecture is known as MapReduce framework. It divides, processes the data, and run it in parallel. However, there are some defects in Hadoop. MapReduce has a high overhead while running a job. Therefore, Hadoop relatively ill-suited while is used for cases of an iterative or low-latency environment [4]. So, there is a need to develop the framework and solve some issues in Hadoop. One of the developed software is Apache Spark that is built on top of Hadoop architecture.

Apache Spark is an open-source distributed computing framework. It is designed to perform the low latency jobs and store in-between data and results in memory. Therefore, it is called Memory Computing that improves the efficiency of data computing. As a result, Spark is better than Hadoop for data mining and machine learning applications [4]. Moreover, to perform scalable machine learning, graph analysis, streaming and structured data processing. There are upper-level libraries in it [5]. The superior feature of Apache Spark is Resilient Distributed Dataset (RDD) which offers fault tolerant objects that can be deployed across a distributed cluster [6]. Spark application have been built to run it there on a small cluster. The advantage of Spark in this case is more about the simplicity of converting the application to a clustered implementation.

Given the tremendous development of data in recent years, especially in the field of healthcare, it became necessary to use a technique that processes, stores and handles such huge data in secure, online system, which led to the introduction of cloud computing. Cloud computing is a technology that connects a group of computers together to provide several services to the user at any time. Cloud computing provides faster data access time, high processing speed, and lower cost [7]. Therefore, the healthcare cloud provides better performance, lower cost, and unlimited scalability [8].

In this research, Amazon EC2 cloud computing model has been used for central part of designed web and for collection and storing of large data Amazon uses S3. Amazon clusters is a group of servers which is working together to perform any

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type of tasks on distributed database on different servers in parallel. The most related works associated with sentiment classification are explained in the following section.

II. Related Works

In 2016, Nodarakis, N. et al present a novel approach for Sentiment Learning within the Spark framework; the proposed algorithm uses the hashtags and emoticons within a tweet as sentiment identifiers, and proceeds to a parallel and distributed classification procedure for various types of sentiment [9].

In 2016, Kharde and Sonawane have compared many of the classification algorithms in Twitter data to perform sentiment analysis and opinion Mining. Performance of NB, Max Entropy, and SVM have been evaluated in the specific domain of tweets. According to the results, SVM model proved to have the best performance [10].

The authors in [11] investigate how MinHash-LSH can help in indexing and searching Web data. They describe MinHash-statistical LSH's tuning technique and compare it to the inverted index in terms of accuracy and performance. They also provide an online demonstration of the index using real-world data.

Baltas A. et al (2017) aim to create an Apache Spark cloud-based Sentiment Analysis platform for Twitter information that classifies tweets using supervised learning techniques. They were experimenting with binary and ternary classification and concentrating on the shift in accuracy induced by the size of the training dataset and the features derived from the feedback [12].

Assefi et al in 2017 have focused on a SVM, Decision Tree, NB, and RF classification algorithms using Apache Spark MLlib 2.0. Then, they have compared the models to the same algorithms in Weka. The outcome shows that Apache Spark Mllib has a robust tool for big data processes [13].

Analyzing the sentiment of diabetes patients on the twitter platform [14].

In 2018, Nair and Shetty developed a health monitoring application based on spark cluster ML model for the prediction of heart disease using Decision Tree methodology to predict an individual's health status by using Twitter real-time data. The software was deployed in "Amazon Elastic Compute Cloud (EC2)" using the cloud [15].

In 2018, V. Setty suggested a distributed architecture for processing news articles and clustering them to make numerous news mining tasks easier. The system's core is a scalable distributed clustering technique based on Locality Sensitive Hashing that is resistant to outliers and noise. He also suggests an online version of the clustering technique to keep the news event clusters continually updated. He uses Apache Spark to accomplish the proposed approach. He demonstrates that the approach beats widely used clustering approaches such as K-Means in both run time and clustering quality using a huge news collection of over 8 million items [16].

Al-Saqqa et al in 2018 collected 4 million Amazon customer review datasets for broad-based sentiment analysis. The dataset was tested for the supervised algorithm of machine-learning, where the model was trained using the labeled training set. It applied classification techniques where vector support outperforms NB and LR, achieving 86 percent accuracy [17].

Singh et al (2018) applied SVM and NB to examine the emotional polarity of health domain tweets. SVM's performance was again observed to be higher accuracy than NB [18].

S. Kumar et al in 2020 Applied SVM and NB to examine the emotional polarity of health domain tweets. SVM's performance was again observed to be higher than NB [19].

In 2020, Yogi, T. N., & Paudel, N. have implemented sentiment analysis which can be used with supervised machine learning for fake positive or negative review identification. SVM algorithm not only passes text detection but can also be used for fake feedback [20].

Based on the advanced strategies introduced in literature review, most of the studies have announced comparable outcomes in terms of accuracy and reliability. Nonetheless, not entirety of the studies is based on the similar datasets. Also, not all studies are implemented in the same environment. Subsequently, they cannot be benchmarked consistently and thought about on a common ground. In the next section, the proposed system and the used data sets are discussed in detail.

III. Proposed Methodology

The main stages of the approach are presented in "Fig. 1". The approach starting with the data preprocessing and feature extraction, then applying the machine learning classifiers: Logistic Regression (LR), Naïve Bayes (NB), Support Vector Machine (SVM) and Random Forest (RF) Classifiers separately under Spark environment in local spark and in YARN spark client modes and in the R traditional system. Finally, the results are evaluated using the accuracy and time metrics and comparing the results with Minhash-LSH algorithm which under spark environment.

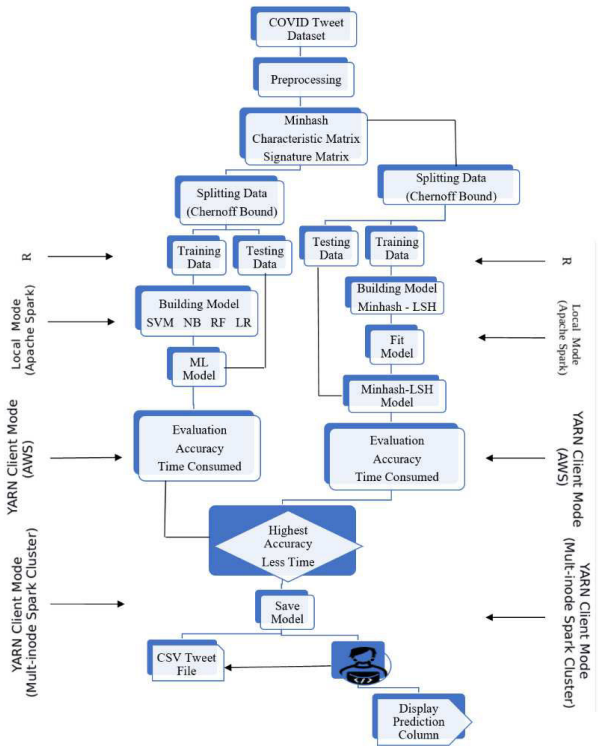

Fig. 1. The Proposed System of Tweets Sentiment Classification

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In short, dataset is CSV file, with a content of plain texts represent tweets, classes, and the ID of each tweet. Each line of the content has a unique ID to refer to it.

Between November 9 and December 8, 2020, of 2349659 tweets concerning the topic of COVID-19 vaccination. Only tweets written in English have been considered. Duplicated tweets have been discarded, as well as retweets. The remaining number of tweets in the cleaned dataset is 752951. From the cleaned dataset they have randomly selected and annotated 7530 tweets, representing approximately 1.00% of all the tweets in the dataset.

The stance of the tweets towards vaccination has been evaluated by three independent human raters into three classes: in favor, against and neutral. Tweets that have been assigned to the class in favor express a positive opinion regarding the vaccination. Tweets belonging to the against vaccination class express a negative opinion towards COVID-19 vaccination. The neutral class mainly includes news related to the development of vaccines, tweets that do not express a clear opinion, such as questions regarding the vaccine, informative tweets concerning vaccination, as well as off-topic tweets. The used balanced annotated dataset is available at the following link: https://github.com/liviucotfas/covid-19-vaccination-stance-detection.

TABLE I. STATISTICS FOR THE USED ANNOTATED DATASET

Class	Number	Precent
Against	789	28.27957
Neutral	1009	36.16487
In favor	992	35.55556
Total	**2790**	**100%**

The second dataset is obtained from the IEEE data port on May 31, 2020. It contains the tweet IDs and sentiment scores of 20765 tweets. Tweets IDs are extracted using the following filters: language “en” and keywords “corona”, “coronavirus”, “covid”, “pandemic” and variants such as “sarscov2”, “nCov”, “covid-19”, “ncov2019”, “2019ncov” and their hashtags. IEEE data port does not provide tweet text, resorting us to develop an in-house crawler that could extract them from Tweeter based on the tweet IDs.

The sentiment scores are classified as positive (greater than 0), neutral (equal to 0) and negative (less than 0). The dataset is split into a training set and a testing set with a ratio of 51:100, respectively. In Table II the statistics of all classes of this dataset.

TABLE II. COUNTS OF TWEETS WITHIN THE THREE CLASSES.

Sentiment	Count
Neutral	7064
Positive	7504
Negative	6197
Total	**20765**

After the data sets downloaded, the next step is data Preprocessing. Given the fact that the tweets from the Twitter portal need a set of processors, the following preprocessing procedures have been adopted to normalize their content. This was done with the tm package in R with its own Tokenizer class:

- Remove punctuations.
- Lowercase and derived letters to reduce word conjugation.
- Remove stop words.
- Removes Hashtags, URLs, Mentions, retweet Favorite (RT-FAV)), Emojis, and Smileys.

There are also several libraries available for cleaning Twitter data sets. These operations were performed rtweet, tidyverse, ggplot2, dplyr, tidytext on all data that was entered into the system to clean it and read the data properly.

Moreover, before using the data by machine learning classifier. There is a need to represent the text in a format that suitable for the classifier to deal with it, so the text is converted into a feature vector. Therefore, the Minhash is used.

Before matching tweets for similarity, they should be represented as groups. This process is called shingling, and it basically constructs the document based on K number of words to generate what is called a “Token”. Shingles for a sentence would be a set created from some consecutive words and then stored into a dictionary.

The k-shingle is K of the tokens (words) seen consecutively in a tweet. Occasionally, to slice short-length strings and use groups of hash values to convey tweets, hash shingles are useful.

The tweets were set to smaller subgroups that were generated from the shingles and were fragmented. The next step in the technology process is to create a characteristic matrix, which represents each group or tweet as columns while indicating the items of standardized groups as rows. Generally, the cell is given a value of 1 when the shingles is part of the group, otherwise it is given a 0.

The Minhash is calculated in the document by specifying change rows for a column from the featured matrix described earlier. As can be seen, the quantity of Minhash for each column is equal to the number of the first row in the alternative request, where the column has a value of 1.

Consequently, the signature matrix consists of the hashes as rows and the tweets as columns. Each column refers to the Minhash signature for the tweet. It should be noticed that the signature matrix is much smaller than the characteristic matrix.

The last step of building the model is the classify phase. Classification is a term used both for the classification method (distinguishing and distributing kinds of” things” into various groups) and for the resulting class collection, as well as for the assigning of items to pre-established classes. Four classification techniques have been used in the proposed method, including NB, SVM, LR and RF. Before input the data to the classifier, the dataset was separated into training (51%) and testing (100%) because of the high rate of accuracy acquired in this division.

Three modes have been used in R, local spark, and in YARN client spark. The number of CPU cores to be assigned to the local operation is either 2 or 4. By using breakpoints while running the code from Rstudio, it aids in debugging. While in Yarn client mode: the driver program runs on the yarn client, and we type the command to submit the spark application (may not be a machine in the yarn cluster). The tasks are completed on the executors in the YARN cluster's node managers in this manner, even though the drive program

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is executing on the client workstation. When using a cluster, we benefit from the Core and RAM of the cluster to perform computations (done by the workers). However, we can decide to run the driver program on the cluster or in the machine.

It has been used the Amazon EMR to create the Apache Spark cluster. There are many features of Amazon EMR for apache spark as below:

The Amazon EMR runtime for Apache Spark can be up to two times faster than clusters that do not use it. It is fully compatible with Apache Spark's standard API. Without making any changes to the applications, this improved performance means that workloads run faster and cost less to compute.

In terms of security, Spark's security is turned off by default. This could imply that we are already vulnerable to attack. Spark provides a variety of deployment modes, each with its own level of security. In all environments, not all deployment types will be secure, and none are secure by default.

Amazon EMR security features can be considered as one reason of choosing it. Customers in regulated areas, such as healthcare, for example, select Amazon EMR as part of their data strategy.

In the local spark mode, the output from minhash function has been entered into the SVM, NB, RF, and LR Classification algorithm, and the performance metrics have been obtained. The user, system, and real time, time consumed, and accuracy are the performance indicators. The time takes for the script to run in real time, user time is the time it takes for the cores to compute, and system time is the time it takes for input/output.

The four machine learning algorithms have been applied to the two datasets (COVID Vaccine Tweet and COVID IEEE Data Port Tweet) in the same way. We repeat the experiment in local R, parallel R, and YARN Client spark mode to obtain the results. The tables listed below show the results of all three modes in results section.

IV. Result

The experiment has repeated in local R, parallel R, and YARN Client mode to obtain the results. The tables listed below show the results of all three modes. Also, the four machine learning algorithms have been applied to the two datasets (COVID Vaccine Tweet and COVID IEEE Data Port Tweet) in the same way.

TABLE III. USER, SYSTEM, AND REAL TIME IN SECOND OF SVM, NB, RF, AND LR ALGORITHMS FOR RESULT FROM MINHASH ALGORITHM ON COVID-19 VACCINE STANCE DATASET

Mode *(k=4, No. of Minhash=20)*	**User Time**	**System Time**	**Real Time**
Local R (SVM)	1006.318	0.069	1006.326
Local Spark (SVM)	1.865	0.459	75.413
YARN Client (SVM)	0.691	0.008	14.662
Local R (NB)	24.586	0.843	25.292
Local Spark (NB)	2.082	0.245	3.32
YARN Client (NB)	1.045	0.012	1.749
Local R (RF)	40.733	0.873	41.664
Local Spark (RF)	2.493	0.305	5.817
YARN Client (RF)	1.103	0.004	2.11
Local R (LR)	17.89	0.003	17.885
Local Spark (LR)	2.598	0.426	17.935
YARN Client (LR)	1.174	0	2.585

Table IV. time consumed of SVM, NB, RF, and LR algorithms for result from Minhash algorithm on COVID-19 Vaccine Stance Dataset (k=4, No. of Minhash=20)

Mode	**SVM**	**NB**	**RF**	**LR**
Local R	17.46788 mins	25.39035 secs	42.15663 secs	18.08444 secs
Local Spark	1.25449 mins	3.877612 secs	8.742572 secs	11.95354 secs

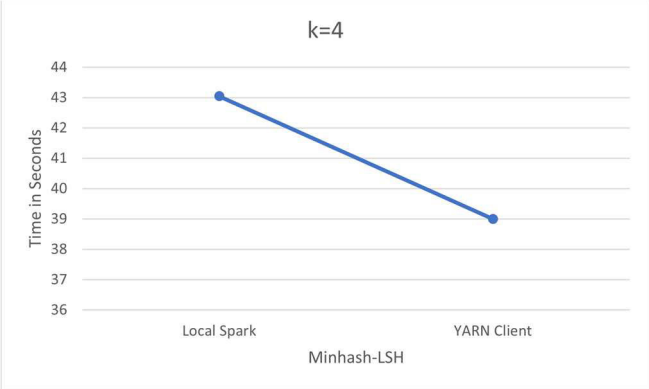

Fig. 2. Performance metric (time consumed) of Minhash-LSH algorithm on COVID-19 Vaccine Stance Dataset (k=4, No. of Minhash=20)

It can be seen in Table III that user, system, and real time by using YARN Client mode are less than in local R and local spark. Moreover, Table IV shows that the R consumed much time than in Apache Spark. Minhash-LSH has consumed much time than Minhash-ML algorithms except Minhash-SVM as shown in Table IV and "Fig. 2'. When k=4, Minhash-NB has consumed lower time than the rest models in local spark mode. While Minhash-LR consumed time is the lowest time in local R.

Table XIII shows that Minhash-LR model accuracy is the most significant value in spark. In the three modes with different number of nodes, the accuracy does not show significant improvement. This is since all modes use the same algorithm. Also, there is no effect on the value of accuracy while using parallel R with 8 cores. The Minhash-MLs models have obtained higher accuracies than Minhash-LSH except Minhash-NB as illustrated in Table XIII.

TABLE V. USER, SYSTEM, AND REAL TIME IN SECOND OF SVM, NB, RF, AND LR ALGORITHMS FOR RESULT FROM MINHASH ALGORITHM (K=5, NO. OF MINHASH=20)

Mode	**User Time**	**System Time**	**Real Time**
Local R (SVM)	210.281	0.039	210.309
Local Spark (SVM)	0.805	0.052	39.846
YARN Client (SVM)	0.434	0.02	12.676
Local R (NB)	2.691	0	2.692
Local Spark (NB)	1.992	0.004	2.707
YARN Client (NB)	1.055	0.004	1.723
Local R (RF)	12.597	0.276	12.871
Local Spark (RF)	2.276	0	3.994

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Mode	User Time	System Time	Real Time
YARN Client (RF)	1.13	0.004	2.119
Local R (LR)	4.872	0	4.866
Local Spark (LR)	2.033	0.016	8.205
YARN Client (LR)	1.159	0.008	2.564

TABLE VI. PERFORMANCE METRIC (TIME CONSUMED) OF SVM, NB, RF, AND LR ALGORITHMS IN LOCAL R AND LOCAL SPARK MODE FOR RESULT FROM MINHASH ALGORITHM (K=5, NO. OF MINHASH=20)

Mode	SVM	NB	RF	LR
Local R	2.839553 mins	2.852424 secs	12.98167 secs	4.981981 secs
Local Spark	43.72447 secs	2.776108 secs	4.124445 secs	8.71395 secs

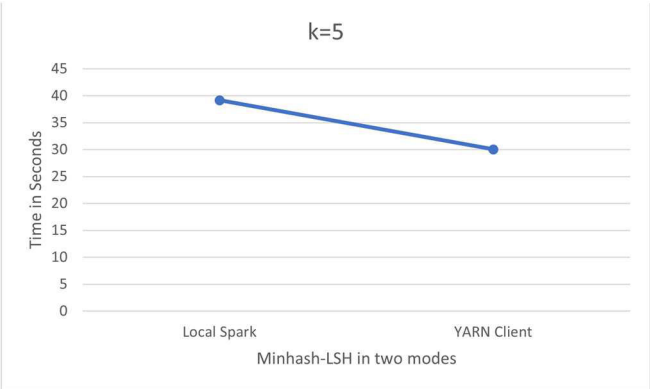

Fig. 3. Performance metric (time consumed) of Minhash-LSH algorithm on COVID-19 Vaccine Stance Dataset (k=5, No. of Minhash=20)

It can be seen in Table V that user and real times by using YARN Client mode are less than in local R and local spark. Moreover, shows that the Minhash-LR consumed in local R less time than in Apache Spark. Minhash-LSH has consumed much time than Minhash-ML algorithms except Minhash-SVM as shown in Table VI and "Fig. 3". Minhash-NB has consumed lower time than the rest models in local R and local spark modes when k=5.

Table XIII shows that the Minhash-MLs models in local R have obtained the highest accuracy in all algorithms with 100%. When they are compared with other modes. Moreover, Minhash-LR model accuracy is the most significant value in local R and YARN Client modes. While in local spark, Minhash-RF accuracy has the highest value. In local spark and YARN Client modes with different number of nodes, the accuracy does not show any improvement. This is since all modes use the same algorithm. Also, there is no effect on the value of accuracy while using parallel R with 8 cores. The Minhash-MLs models have obtained higher accuracies than Minhash-LSH except Minhash-NB as illustrated in Table XIII.

Number of shingles has been chosen as k=4 or k=5. When k=5, RF accuracy is better than when k=4 in all modes. In contrast, Minhash-NB accuracy is the best if k=4 in all modes. K=5 is the best choice in Minhash-SVM, Minhash-RF, and Minhash-LR models in local R and YARN Client mode and in Minhash-LSH model. While k=4 is the best option in Minhash-NB model in all modes and Minhash-SVM and Minhash-LR in local spark mode.

TABLE VII. USER, SYSTEM, AND REAL TIME IN SECOND OF SVM, NB, RF, AND LR ALGORITHMS FOR RESULT FROM MINHASH ALGORITHM ON COVID IEEE DATA PORT TWEET DATASET (K=4, NO. OF MINHASH=20)

Mode	User Time	System Time	Real Time
Local R (SVM)	388.950	0.000	388.921
Local Spark (SVM)	1.106	0.257	38.806
YARN Client (SVM)	0.732	0.044	16.521
Local R (NB)	7.969	0.268	8.188
Local Spark (NB)	2.287	0.287	3.378
YARN Client (NB)	1.374	0.008	2.160
Local R (RF)	58.330	0.345	58.680
Local Spark (RF)	2.693	0.372	5.389
YARN Client (RF)	1.459	0.004	3.212
Local R (LR)	6.561	0.000	6.556
Local Spark (LR)	2.601	0.255	8.970
YARN Client (LR)	1.487	0.004	3.657

TABLE VIII. PERFORMANCE METRIC (TIME CONSUMED) OF SVM, NB, RF, AND LR ALGORITHMS IN LOCAL R AND LOCAL SPARK MODE FOR RESULT FROM MINHASH ALGORITHM ON COVID IEEE DATA PORT TWEET DATASET (K=4, NO. OF MINHASH=20)

Mode	SVM	NB	RF	LR
Local R	6.416187 mins	9.001455 secs	1.053418 mins	6.585283 secs
Local Spark	40.55874 secs	3.458827 secs	5.510594 secs	8.947233 secs

Minhash-NB has consumed lower time than the rest models in local spark mode when k=4 with 3.459 seconds. 6.585 seconds have been spent by performing Minhash-LR model in local R mode. It is the lowest time in that mode as shown in Table VIII.

TABLE IX. PERFORMANCE METRIC (TIME CONSUMED) OF MINHASH-LSH ALGORITHM (K=4, NO. OF MINHASH=20)

Minhash-LSH	k=4
Local Spark	55.40 secs
YARN Client	49.568 secs

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TABLE X. USER, SYSTEM, AND REAL TIME OF SVM, NB, RF, AND LR ALGORITHMS FOR RESULT FROM MINHASH ALGORITHM ON COVID IEEE DATA PORT TWEET DATASET (K=5, NO. OF MINHASH=20)

Mode	User Time	System Time	Real Time
Local R (SVM)	178.842	0.018	178.841
Local Spark (SVM)	0.552	0.028	23.994
YARN Client (SVM)	0.697	0.016	16.837
Local R (NB)	9.360	0.048	9.411
Local Spark (NB)	1.824	0.004	2.693
YARN Client (NB)	1.354	0.012	2.137
Local R (RF)	40.422	0.413	40.835
Local Spark (RF)	2.017	0.004	4.112
YARN Client (RF)	1.459	0.004	2.766
Local R (LR)	6.677	0.004	6.677
Local Spark (LR)	1.827	0.012	7.645
YARN Client (LR)	1.542	0.016	3.395

TABLE XI. PERFORMANCE METRIC (TIME CONSUMED) OF SVM, NB, RF, AND LR ALGORITHMS IN IN LOCAL R AND LOCAL SPARK MODES FOR RESULT FROM MINHASH ALGORITHM ON COVID IEEE DATA PORT TWEET DATASET (K=5, NO. OF MINHASH=20)

Mode	SVM	NB	RF	LR
Local R	3.798173 mins	11.26488 secs	40.55104 secs	6.592923 secs
Local Spark	26.33183 secs	2.660438 secs	4.04249 secs	7.561792 secs

TABLE XII. PERFORMANCE METRIC (TIME CONSUMED) OF MINHASH-LSH ON COVID IEEE DATA PORT TWEET DATASET (K=5, NO. OF MINHASH=20)

Minhash-LSH	k=5
Local Spark	30.53 secs
YARN Client	26.652 secs

V. ANALYSIS AND DISCUSSION

In comparison to YARN Client mode, the algorithms in Local R mode take a lot longer. All the algorithms can be run in parallel, resulting in significant time savings. When we have compared among all four algorithms in local R, Minhash-SVM and Minhash-RF have taken much time than the rest especially Minhash-NB. Overall, the YARN Client mode takes significantly less time to run algorithms than the local spark mode, proving that parallel computing with spark uses more cores and reduces processing time.

Our model's result can be compared with another research which have used the same dataset. Therefore, authors in [21] have used the same COVID-19 Vaccine Stance dataset and found the results as in the Table XIII. Therefore, we have compared our results with the accuracy in Table XIII and found that our results Minhash-ML except Minhash-NB are better than their algorithms' results. Minhash-LSH accuracy in our results is about 0.84.

TABLE XIII. THE ACCURACY OF SOME USED ALGORITHMS IN [21] THAT USED THE COVID-19 VACCINE STANCE DATASET.

Method	Accuracy
(1,2) MNB n-gram ; 3000 Features	0.74
(1,2) MNB n-gram ; 2000 Features	0.73
(1,1) RF n-gram ; all Features	0.72
(1,2) RF n-gram ; 2000 Features	0.71
(1,2) SVM n-gram ; all Features	0.76
(1,2) SVM n-gram ; 2000 Features	0.74
CNN Word embedding: Fasttext.wiki-news-300d	0.69
Bidirectional Encoder Representations from Transformers (BERT)	0.79
Our Mode Minhash-LSH (K=4)	0.836
Our Mode Minhash-SVM (K=4)	0.9805015
Our Mode Minhash-NB (K=4)	0.5222
Our Mode Minhash-RF (K=4)	0.9861
Our Mode Minhash-LR (K=4)	0.9916
Our Mode Minhash-LSH (K=5)	0.841
Our Mode Minhash-SVM (K=5)	0.9836601
Our Mode Minhash-NB (K=5)	0.5082967
Our Mode Minhash-RF (K=5)	0.9902
Our Mode Minhash-LR (K=5)	1

Authors in [22] have used the same COVID IEEE Data Port Tweet dataset and found the results as in the Table XIV. Therefore, we have found that our results of Minhash-MLs models except Minhash-NB are better than their algorithms' results in terms of accuracy. However, Minhash-LSH accuracy in our results is less than 0.86 as in Table XIV.

TABLE XIV. THE ACCURACY OF SOME USED ALGORITHMS IN [22] THAT USED THE COVID IEEE DATA PORT TWEET DATASET.

Approach/Model	Accuracy
(TF-TDF) – RF	0.86
(TF-TDF)-XGboost	0.85
(TF-TDF)-SVC	0.85
(TF-TDF)-ETC	0.88
(TF-TDF)-DT	0.83
BoW-RF	0.87
BoW-XGboost	0.86
BoW-SVC	0.82
BoW-ETC	0.88
BoW-DT	0.85
(TF-IDF and BoW) – RF	0.88
(TF-IDF and BoW) – Xgboost	0.89
(TF-IDF and BoW) – SVC	0.86
(TF-IDF and BoW) – ETC	0.89
(TF-IDF and BoW) – DT	0.82
Our Mode Minhash-LSH (K=4)	0.8500
Our Mode Minhash-SVM (K=4)	1
Our Mode Minhash-NB (K=4)	0.5072333
Our Mode Minhash-RF (K=4)	1
Our Mode Minhash-LR (K=4)	1
Our Mode Minhash-LSH (K=5)	0.8543
Our Mode Minhash-SVM (K=5)	1
Our Mode Minhash-NB (K=5)	0.5109733
Our Mode Minhash-RF (K=5)	1
Our Mode Minhash-LR (K=5)	1

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VI. Conclusions and Future Works

To conclude, the distributed and parallel computing, both modes use the set number of cores. This is evidenced by the higher user time even though the real execution time is substantially shorter. The employment of cores for parallel computation is shown by increased user time. Moreover, the usage of a larger number of cores in the Local Cluster mode reduces the algorithm's execution time slightly, but not much. This is due to the Local mode, which uses a single machine to create the illusion of parallelism.

The running time of algorithms in the Hadoop YARN Client mode is faster than the local spark mode. Other performance measurements, such as accuracy, indicate no discernible differences between the two modes. This shows that the internal implementation of machine learning algorithms is unaffected by the number of cores used. Although the predictions are computed faster, the method by which they are produced stays the same and produces nearly identical results every time.

K-shingles help to recode data set into actual class assignment which lead to a high accuracy result compared with standard works. Minhash technique has increased the quality of the whole work, it used to distribute the tweets in iterative process to generate characteristics matrix and signature matrix. When compared with the original dataset, notice a very clear difference in the increase in the accuracy of classification when using data mining classifier with Minhash algorithm.

Data mining classifier (SVM – RF – NB - LR) are chosen because have a higher accuracy, in comparison with other classifier that used it in related work, in this dissertation LR and RF are the best algorithms. According to the results obtained, Spark has a good speed in doing data processing when the dataset is large. Moreover, the hybrid methods (Minhash with MLs) have obtained best accuracy when compared with other studies.

As future work, further experiments can be conducted to collect more tweets. Also, we hope the collected tweets have better sentence structure and words to get a more valid accuracy of data values. We can implement more classification models with different feature sets and n-gram models (bi-gram and tri-gram) that may enhance the performance of the classification. This model can be used to discover sentiment classification within an Online data stream on social media networks by make integration with API twitter. Moreover, production-quality classifier will incorporate many different features beyond the vectors corresponding to the words in the text. Meaning that add feature like location of user, age of user and node connected link to other user this may be help us to make it feature selection to improve the classification performance.

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